

An Analysis of Fish Data in the North East Texas and South East Oklahoma

In this file I have combined all of the scripts, functions, and figures that I have found insightful or relevant towards the interpretation of this data.

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Importing all of the Necessary Libraries

As I write this code, I will import **ALL** of the libraries I use in this cell

```
In [1]: import pandas as pd # library for data
import numpy as np # library for vecto
import matplotlib.pyplot as plt # library for data
import seaborn as sns # library for data
from sklearn.model_selection import train_test_split # library for split
from python_functions.label_plot_edges import label_plot_edges # function to label
```

Loading the Files as Pandas Data Frames

File	AS	Variable
gov_envData10dts_2025.csv	as	g_ed
gov_envData10dtsMetrics_2026.csv	as	g_edm
seth_ecoRegionData_2025.csv	as	s_erd
seth_envData_2025.csv	as	s_ed
seth_envGenusCountData_2025.csv	as	s_egcd
seth_fishCountData_2025.csv	as	s_fcd
seth_genusCountData_2025.csv	as	s_gcd

File	AS	Variable
seth_gov_envData10dtsMetrics_genusCount2026.csv	as	sg_edmgc

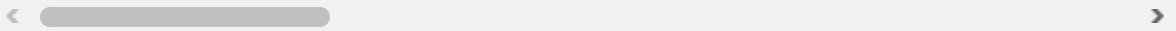
```
In [2]: g_ed = pd.read_csv('../data/gov_envData10dts_2025.csv')
g_edm = pd.read_csv('../data/gov_envData10dtsMetrics_2026.csv')
s_erd = pd.read_csv('../data/seth_ecoRegionData_2025.csv')
s_ed = pd.read_csv('../data/seth_envData_2025.csv')
s_egcd = pd.read_csv('../data/seth_envGenusCountData_2025.csv')
s_fcd = pd.read_csv('../data/seth_fishCountData_2025.csv')
s_gcd = pd.read_csv('../data/seth_genusCountData_2025.csv')
sg_edmgc = pd.read_csv('../data/seth_gov_envData10dtsMetrics_genusCount2026.csv')
```

```
In [3]: # g_ed.info()      # check the structure of the data frame
g_ed.head()      # printing the first 5 rows of the data frame
```

```
Out[3]:
```

	Segment_10Day	site_0.burn	site_0.dfm1000hr	site_0.dfm100hr	site_0.energy_ERC	site_0
0	0	23.4	20.15	21.12	18.1	
1	1	25.5	19.36	17.29	24.5	
2	2	25.3	18.09	15.66	28.3	
3	3	30.9	17.65	16.91	29.1	
4	4	16.2	17.47	17.54	23.8	

5 rows × 829 columns

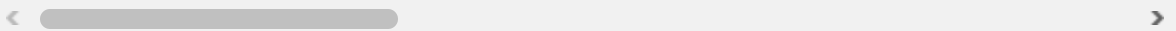


```
In [4]: # g_edm.info()    # check the structure of the data frame
g_edm.head()    # printing the first 5 rows of the data frame
```

```
Out[4]:
```

	site	burn_mean	burn_median	burn_sd	burn_min	burn_max	burn_range	burn_cv
0	site_1	24.772945	24.40	8.686572	5.7	57.2	51.5	0.350648
1	site_2	24.673630	24.25	8.591869	4.3	57.3	53.0	0.348221
2	site_3	25.542466	25.20	8.994322	3.0	59.1	56.1	0.352132
3	site_4	26.318836	26.30	9.289094	2.7	62.3	59.6	0.352945
4	site_5	24.677854	24.30	8.547560	5.5	55.3	49.8	0.346366

5 rows × 145 columns



```
In [5]: # s_erd.info()    # check the structure of the data frame
s_erd.head()    # printing the first 5 rows of the data frame
```

Out[5]:

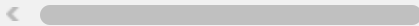
	Site_N	eco_Number	eco_Letter	Title	State
0	1	33	a	Northern Post Oak Savana	TX
1	2	33	a	Northern Post Oak Savana	TX
2	3	33	a	Northern Post Oak Savana	TX
3	4	32	a	Northern Black Land Prairie	TX
4	5	35	c	Pleistocene Fluvial Terraces	TX

In [6]: `# s_ed.info()` *# check the structure of the data frame*
`s_ed.head()` *# printing the first 5 rows of the data frame*

Out[6]:

	SiteN	State	Lat	Long	DO	pH	avg_WW	avg_Depth	pf_Pool	pf_
0	1	TX	33.741217	-95.927376	10.7	6.99	5.76666	23.75000	0.000000	83.333
1	2	TX	33.717896	-95.911323	4.2	7.22	2.94000	27.00000	60.000000	20.000
2	3	TX	33.747466	-96.019146	7.4	7.58	11.21666	46.33333	66.666667	33.333
3	4	TX	33.554952	-96.169601	7.7	7.86	10.51666	18.41666	0.000000	66.666
4	5	TX	33.823594	-95.861067	6.8	7.42	22.00000	39.66666	100.000000	0.000

5 rows × 26 columns



In [7]: `# s_egcd.info()` *# check the structure of the data frame*
`s_egcd.head()` *# printing the first 5 rows of the data frame*

Out[7]:

	SiteN	State	Lat	Long	DO	pH	avg_WW	avg_Depth	pflow_Pool	pflow
0	1	TX	33.741217	-95.927376	10.7	6.99	5.76666	23.75000	0.000000	83.333
1	2	TX	33.717896	-95.911323	4.2	7.22	2.94000	27.00000	60.000000	20.000
2	3	TX	33.747466	-96.019146	7.4	7.58	11.21666	46.33333	66.666667	33.333
3	4	TX	33.554952	-96.169601	7.7	7.86	10.51666	18.41666	0.000000	66.666
4	5	TX	33.823594	-95.861067	6.8	7.42	22.00000	39.66666	100.000000	0.000

5 rows × 58 columns

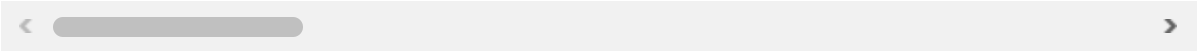


In [8]: `# s_fcd.info()` *# check the structure of the data frame*
`s_fcd.head()` *# printing the first 5 rows of the data frame*

Out[8]:

	SiteN	State	Etheostoma_radiosum	Lepomis_megalotis	Lepomis_macrochirus	Lepomis_g...
0	1	1	0	6	5	
1	2	1	0	0	0	
2	3	1	0	24	12	
3	4	1	0	43	54	
4	5	1	0	16	16	

5 rows × 54 columns

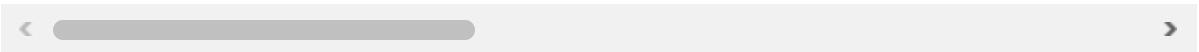


```
In [9]: # s_gcd.info()      # check the structure of the data frame
s_gcd.head()           # printing the first 5 rows of the data frame
```

Out[9]:

	SiteN	State	Ameiurus	Aphredoderus	Aplodinotus	Campostoma	Cyprinella	Cyprinus
0	1	TX	0	0	0	1	0	0
1	2	TX	0	0	0	0	0	0
2	3	TX	0	0	0	0	0	0
3	4	TX	0	0	0	36	39	1
4	5	TX	0	0	1	0	22	0

5 rows × 32 columns

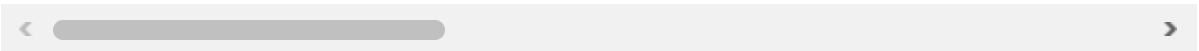


```
In [10]: # sg_edmgc.info()    # check the structure of the data frame
sg_edmgc.head()              # printing the first 5 rows of the data frame
```

Out[10]:

	SiteN	State	Lat	Long	DO	pH	avg_WW	avg_Depth	pflow_Pool	pflow
0	1	TX	33.741217	-95.927376	10.7	6.99	5.76666	23.75000	0.000000	83.33
1	2	TX	33.717896	-95.911323	4.2	7.22	2.94000	27.00000	60.000000	20.00
2	3	TX	33.747466	-96.019146	7.4	7.58	11.21666	46.33333	66.666667	33.33
3	4	TX	33.554952	-96.169601	7.7	7.86	10.51666	18.41666	0.000000	66.66
4	5	TX	33.823594	-95.861067	6.8	7.42	22.00000	39.66666	100.000000	0.00

5 rows × 202 columns



Species, Genus, Site Attributes, and Site Metrics Bar Plots

In this Section I will generate encompassing bar plots to get good overviews of the # of fish species, genus, site attributes and metrics.

I will hope to encompass any relevant but overwhelming data into basic bar plots to get a simple start on understanding this data visually and identify any trends or under-represented features.

Fish Species Bar Plot

This cell simply sums all of the fish species columns in the **seth_fishCountData_2025.csv** dataset and visualizes it as a bar plot.

Here I am looking to see the distribution of species so that I can determine species is abundance and few. My hope is that this will be useful in determining outliers later, and possibly visualize species in fewer numbers than expected.

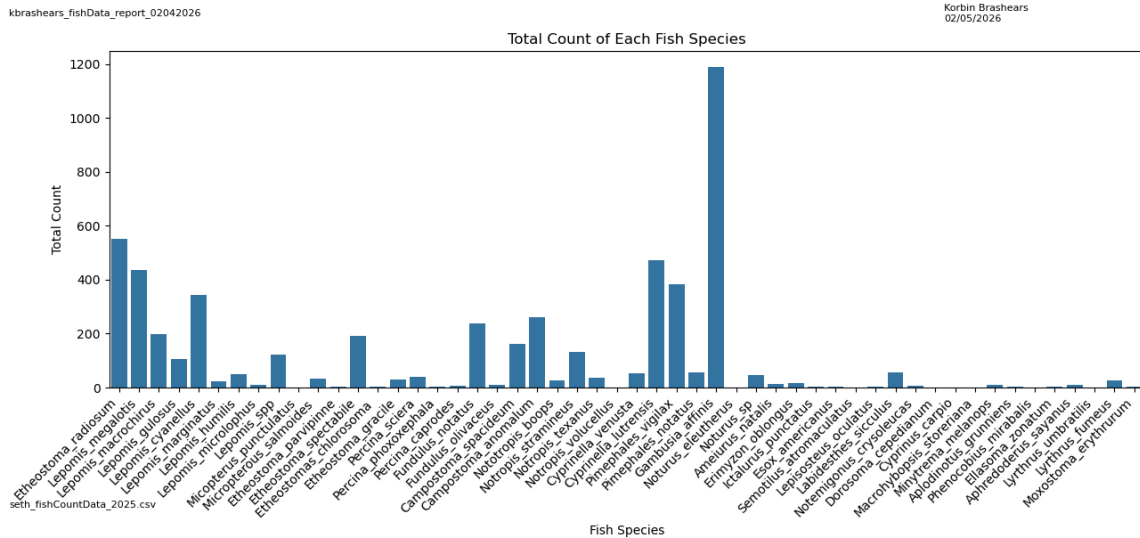
```
In [11]: s_fcd_species_cols = s_fcd.columns[2:]           # selecting
fish_species_sum = s_fcd[s_fcd_species_cols].sum()       # summing e

plt.figure(figsize=(15, 6))                             # creating
sns.barplot(x=fish_species_sum.index, y=fish_species_sum.values) # creating
plt.xticks(rotation=45, ha='right')                     # rotating
plt.xlabel('Fish Species')                              # setting x
plt.ylabel('Total Count')                               # setting y
plt.title('Total Count of Each Fish Species')           # setting t

plt.subplots_adjust(bottom=0.25)                         # adjusting

label_plot_edges(
    program_names="kbrashears_fishData_report_02042026", # calling t
    author_names="Korbin Brashears",
    data_filenames="seth_fishCountData_2025.csv",
    save=False
)

plt.show()                                              # displayin
```



Fish Genus Bar Plot

This cell preforms the same action as before on the **seth_genusCountData_2025.csv**, summing the columns and visualizing as a bar plot.

Hopefully this can help to find outliers with unexpected differences between the population in an over all genus vs a species.

```
In [12]: s_gcd_genus_cols = s_gcd.columns[2:] # selecting
fish_genus_sum = s_gcd[s_gcd_genus_cols].sum() # summing e

plt.figure(figsize=(15, 6)) # creating
sns.barplot(x=fish_genus_sum.index, y=fish_genus_sum.values) # creating
plt.xticks(rotation=45, ha='right') # rotating
plt.xlabel('Fish Genus') # setting x
plt.ylabel('Total Count') # setting y
plt.title('Total Count of Each Fish Genus') # setting t

plt.subplots_adjust(bottom=0.25) # adjusting

label_plot_edges(
    program_names="kbrashears_fishData_report_02042026", # calling t
    author_names="Korbin Brashears",
    data_filenames="seth_genusCountData_2025.csv",
    save=False
)

plt.show() # displayin
```

